Radiation Oncology Fundamentals

Radiobiology, Treatment Planning, Techniques, and Toxicity

Doc #06 of 15	**Domain:** Oncology & Cancer Treatment	**Audience:** Medical Professionals

Keywords: radiotherapy | IMRT | SBRT | fractionation | radiobiology | brachytherapy | proton

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1. Radiobiology — The 5 R's

R	Full Term	Clinical Implication
Repair	Repair of sub-lethal damage	Normal tissues repair between fractions; tumours less so → fractionation benefit
Redistribution	Cell-cycle redistribution	Cells in radioresistant S-phase move to sensitive G2/M between fractions
Repopulation	Clonogen repopulation	Tumour cells repopulate during prolonged courses → accelerated fractionation rationale
Reoxygenation	Tumour reoxygenation	Hypoxic (radioresistant) cells reoxygenate as outer layers are killed
Radiosensitivity	Intrinsic sensitivity	Determines α/β ratio; low α/β (late-reacting) tissues damaged more by large fractions

2. Dose & Fractionation Principles

The linear-quadratic (LQ) model describes cell survival: $S = \exp(-\alpha d - \beta d^2)$, where d is dose per fraction. The α/β ratio distinguishes early-reacting (α/β ~10 Gy) from late-reacting tissues (α/β ~3 Gy). Biologically Equivalent Dose (BED) = $nd \times (1 + d/(\alpha/\beta))$ normalises different schedules for comparison.

Schedule	Dose/Fraction	# Fractions	Total Dose	BED (α/β=10)
Conventional	1.8-2 Gy	25-33	45-66 Gy	53-79 Gy10
Hypofractionation (breast)	2.67 Gy	15	40.05 Gy	50.8 Gy10
Ultra-hypofractionation (prostate)	7.25 Gy	5	36.25 Gy	62.5 Gy10
SBRT (lung NSCLC)	10-20 Gy	3-5	45-60 Gy	~100-180 Gy10
SRS (brain met)	15-24 Gy	1	15-24 Gy	37-82 Gy10

3. Modern RT Techniques

- 3D-CRT — CT-based planning with field shaping; replaced 2D simulation
- IMRT (Intensity-Modulated RT) — computer-optimised multi-leaf collimator modulation; reduces dose to OARs
- VMAT (Volumetric Modulated Arc Therapy) — continuous gantry rotation with dynamic MLC; faster delivery than IMRT
- IGRT (Image-Guided RT) — kV/CBCT or MR-Linac daily imaging; accounts for set-up uncertainty and organ motion
- SBRT / SABR — high ablative doses in 1-5 fractions; lung, liver, spine, oligomets
- SRS — single-fraction radiosurgery for intracranial targets (Gamma Knife, CyberKnife, Linac-SRS)
- Proton therapy — Bragg peak delivers dose conformally with minimal exit dose; CNS, paediatric, re-irradiation
- Brachytherapy (HDR/LDR) — intracavitary (cervix, endometrium) or interstitial (prostate, breast) radiation

4. Acute & Late Toxicities by Site

Treatment Site	Acute Toxicities	Late Toxicities	Dose-Limiting OAR
Head & Neck	Mucositis, dysphagia, xerostomia	Fibrosis, osteoradionecrosis, xerostomia (permanent)	Parotid Dmean <26 Gy
Thorax / Lung	Oesophagitis, fatigue	Radiation pneumonitis, pericarditis	Mean lung dose <20 Gy
Pelvis (prostate)	Proctitis, dysuria	Haematuria, rectal bleeding, ED	Rectal V75 <15%
Pelvis (cervix)	Diarrhoea, cystitis	Vaginal stenosis, fistula, bowel obstruction	Bowel V45 <195cc
Brain	Fatigue, alopecia, nausea	Neurocognitive decline, radionecrosis	Hippocampus Dmean <9 Gy

■ **DOSE CONSTRAINTS: Organ-at-risk (OAR) dose constraints are technique-, fractionation-, and institution-specific. Always refer to current protocol or validated constraint library (e.g., QUANTEC, TG-101).**